Z-J-1843
April 10, 1974

FEDERAL SPECIFICATION

JUICE, LEMON, RECONSTITUTED, PACKET, INDIVIDUAL

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 This specification provides for a reconstituted lemon juice in individual packets as an item of general issue, for all Federal agencies.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

L-C-110	-	Cellophane (Coated and Noncoated Regenerated Cellophane Film)
PPP-B-636	-	Boxes, Shipping, Fiberboard

Federal Standard:

FED-STD-123	-	Marking for Domestic Shipment (Civil Agencies)

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

FSC 8915

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(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-668	-	Sanitary Standards for Food Plants

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

PUBLICATIONS

U.S. Department of Health, Education and Welfare

Federal Food, Drug, and Cosmetic Act.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

U.S. Department of Agriculture

U.S. Standard for Condition of Food Containers

(Application for copies should be addressed to the Information Division, Agricultural Marketing Service, U.S. Department of Agriculture, Washington, DC 20250.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

Association of Official Analytical Chemists

Official Methods of Analysis of the Association of Official Analytical Chemists

(Application for copies should be addressed to the Association of Official Analytical Chemists, Box 540, Benjamin Franklin Station, Washington, DC 20044.)

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National Research Council

Food Chemicals Codex

(Application for copies should be addressed to the Office of Director, National Academy of Science, National Research Council, Washington, DC 20418.)

National Motor Freight Traffic Association, Inc., Agent

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Materials. All material shall be clean, sound and wholesome and free from evidence of insect infestation or other objectionable foreign matter.

3.1.1 Frozen concentrated lemon juice. The frozen concentrated lemon juice used for reconstitution shall be prepared from unfermented juice obtained from mature lemons of one or more of the high acid varieties. Preparation and concentration of the juice shall be in accordance with methods which will insure maximum preservation of its natural characteristics.

3.1.2 Chemical preservative. The chemical preservative(s) added shall be of the types and quantities approved by the Food and Drug Administration and shall conform to Food Chemicals Codex.

3.2 Finished product. The reconstituted lemon juice shall have good color that is typical of lemon juice, practically free of browning caused by scorching, oxidation, storage conditions or other causes. The juice shall have a distinct

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characteristic lemon odor and flavor free from terpene-like, oxidized, scorched, carmelized or other undesirable flavors. The juice shall contain no seeds or portions thereof, no coagulated pulp, peel, core, seed particles or other defects that detract from the appearance or utility of the product. The acid content, calculated as anhydrous citric acid shall be not less than 5.1 grams per 100 ml of reconstituted lemon juice when tested in accordance with 4.4, (each 4 gram) (0.13 fluid ounce packet approximates the juice of one-quarter of a fresh lemon). Storage life at 50°F is 120 days.

3.2.1 Storage. The packaged product shall be maintained at a temperature of 50°F or less.

3.3 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

3.4 Workmanship. The product shall be prepared, processed, and packaged under modern sanitary conditions and in accordance with good commercial practice. For Military agencies the product shall be processed in establishments meeting the requirements of MIL-STD-668.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Preacceptance inspection (plant sanitation inspection).

4.2.1 Civil agencies. When specified (see 6.1), inspection of plants producing the product will be conducted upon request of the contracting officer to the appropriate Government inspection agency. Award of contracts will be limited to establishments known to maintain proper sanitary conditions or which have received prior sanitary approval recommended by the inspection agency.

4.2.2 Military agencies. The product shall be rejected if produced in plants not meeting the requirements of 3.4.

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4.3 Acceptance inspection.

4.3.1 Condition inspection of containers. When inspection of product for quality and condition is performed by the U.S. Department of Agriculture, the United States Standards for Condition of Food Containers shall apply. (For applicable criteria governing inspection of containers for Military agencies, see 4.3.4).

4.3.2 Sampling for inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except as otherwise indicated herein.

4.3.3 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase documents.

4.3.3.1 Preservatives. Conformance of preservatives to Food Chemicals Codex shall be ascertained by examination of pertinent labels on the containers of preservatives. Use of preservatives not conforming to Food Chemicals Codex shall be cause for rejection of the involved finished product.

4.3.3.2 Pouches. Conformance of pouch material to 5.1.1 shall be ascertained by examination of pertinent labels, invoices or other valid documents.

4.3.4 Examination of finished product. Examinations shall be in accordance with table II through V and 4.3.4.1 through 4.3.4.3. Sampling plans for the aforementioned tables shall be in accordance with table I. The acceptable quality levels (AQLs) shall be expressed as percent defective for tables II, III, and IV, and defects per hundred units for table V.

TABLE I - Sampling plans

Table	Sample unit	Lot size expressed in	Inspection level	AQLs	
				Major	Minor
II	1 filled and sealed pouch	Pouches	S-3	-	2.5
III	1 filled and sealed pouch	Pouches	S-3	0.65	-
IV	1 filled intermediate carton	Cartons	S-3	-	2.5
V	1 filled and sealed pouch	Pouches	S-2	1.5	-

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TABLE II - Examination of net weight 1/

Category	Defect
Minor	
201	Less than 3.8 grams. 2/

1/ Weighing device shall be calibrated in no larger than 0.1 gram increments.

2/ The lot shall be rejected if the average net weight of the sample is less than the net weight specified.

TABLE III - Examination for leakage 1/

Category	Defect
Major	
101	Leaker.

1/ See 4.4 for procedure.

TABLE IV - Examination for count of pouches

Category	Defect
Minor	
201	Less than 144 pouches.

TABLE V - Examination of product 1/ 2/

Category	Defect
Major	
101	Color not as required.
102	Presence of seed, peel or core particles or coagulated pulp, or other harmless extraneous vegetable material.

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TABLE V - Examination of product 1/ 2/ (cont'd)

Category	Defect
Major	
103	Not a characteristic lemon flavor and odor.
104	Presence of terpene like, oxidized, scorched, carmelized or other undesirable flavor.

1/ Condition product to 65-75°F.

2/ Presence of foreign material (e.g., filth, metal, glass, wood, paint, insects, etc.) shall be cause for rejection of the lot.

4.3.4.1 Examination of pouches. Examination of pouches for condition and labeling shall be in accordance with the criteria in the United States Standards for Condition of Food Containers.

4.3.4.2 Examination of shipping containers. The filled and closed shipping containers shall be examined in accordance with the quality assurance provision in the appendix of PPP-B-636. In addition, the following defects shall be included in the examination: Major - markings missing, incorrect or illegible; Minor - number of interior cartons not as marked; Minor - markings indicate a total net weight greater than 20 pounds; Minor - labeling on cartons missing or not as required. For level C pack, only the aforementioned defects pertinent to markings, total net weight, labeling and number of interior cartons shall apply.

4.3.4.3 Examination of temperature. Conformance to temperature requirements of 3.2.1 shall be ascertained by examination of temperature recordings in storage areas. Nonconformance shall be cause for rejection of the involved finished product.

4.3.5 Testing of finished product. The finished product shall be tested for acid content (calculated as anhydrous citric) in accordance with 4.4. The sample for testing shall be a 4-ounce composite derived from an adequate number of pouches randomly selected. Lot size shall be expressed in pouches. Test result shall be reported to the nearest 0.1 gram. A test result of less than 5.1 grams of anhydrous citric acid per 100 ml of reconstituted lemon juice shall be cause for rejection of the lot.

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4.4 Examination and test methods. Examination and test procedures which differ from those specified herein, unless otherwise excepted, may be used by the supplier if they provide a quality assurance equivalent to that specified. If the Government contracting officer determines that such procedures and controls do not provide, as a minimum, such quality assurance, the supplier will use the procedures set forth herein. In case of dispute as to examination or test results, the procedures specified herein will govern.

4.4.1 Examination procedures.

4.4.1.1 Examination for leakage of pouch. Pouch seals shall be examined by submerging the pouch in a vacuum desiccator or similar device and drawing a vacuum of 10 inches (atmospheric pressure 29.9 inches) for 10 seconds and observing for leaks. Leakage is indicated by large quantities of air coming from any point in the pouch or a fine stream of bubbles anywhere in the seal area.

4.4.2 Test procedures.

4.4.2.1 Chemical analysis. Chemical analysis shall be made in accordance with the following methods of the Official Methods of Analysis of the Official Analytical Chemists:

Chapter	Test and Method
Fruits & Fruit Products	Titratable acidity* Indicator method:

* Except that results shall be reported as grams of citric acid per 100 ml.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Reconstituted lemon juice shall be packaged in accordance with level B or C, as specified (see 6.1).

5.1.1 Level B. Four grams (0.13 fluid ounce) of reconstituted lemon juice shall be packaged in a pouch sealed on four sides. A minus .2-gram tolerance will be allowed in any one pouch, provided the average net weight of any pouch inspected in table II is not less than 4 grams. The pouch shall be constructed from 0.002-inch thick, white polyethylene, laminated or hot extruded to cellophane complying with type VII, class 1 of L-C-110. The pouch shall be provided with a tear nick near one end to facilitate opening of the pouch. The pouch shall not leak when examined in accordance with 4.4.1.1.

5.1.1.1 Intermediate carton. One hundred and forty-four pouches of the reconstituted lemon juice shall be packaged in a folding paperboard box.

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5.1.2 Level C. Reconstituted lemon juice, in the quantity specified in 5.1.1, shall be packaged in a manner to protect it from deterioration or damage during shipment from the supply source to the first receiving activity. One hundred and forty-four pouches shall be packaged in an intermediate folding paperboard box. The supplier may use his standard practice when it meets these requirements.

5.2 Packing. Not more than 20 pounds of reconstituted lemon juice in intermediate cartons shall be packed in a shipping container in accordance with level B or C, as specified (see 6.1).

5.2.1 Level B. The shipping container shall be a fiberboard box, constructed, closed, and reinforced in accordance with style RSC, W5c, or W5s, of PPP-B-636.

5.2.1.1 When specified (see 6.1 and 6.3), the shipping container specified in 5.2.1 shall be reinforced with nonmetallic strapping or pressure-sensitive adhesive, filament reinforced tape in accordance with the appendix of PPP-B-636, except that two reinforcing bands shall be used, one lengthwise and one girthwise.

5.2.2 Level C. The shipping container shall be in accordance with Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

5.3 Labeling and marking.

5.3.1 Pouches. The pouches shall be labeled with the name of the product, the net weight or volume, and the name and address of the supplier. Reverse printing on the cellophane shall be used.

5.3.2 Intermediate cartons. Labeling for intermediate cartons shall includ identification of contents and the count.

5.3.3 Shipping containers (military agencies). Shipping containers shall be marked in accordance with MIL-STD-129, and the marking shall include the following precautions:

KEEP REFRIGERATED

(50°F. or BELOW)

Inspection/Test Date - DAY MONTH YEAR

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5.3.4 Shipping containers (civil agencies). Shipping containers shall be marked in accordance with the requirements of FED-STD-123 and such other information as specified by individual agencies. In addition, the precautionary markings specified in 5.3.3 shall be included on the shipping container.

6. NOTES:

6.1 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number and date of this specification.
(b) When plant sanitation inspection shall be performed by Civil agencies (see 4.2.1).
(c) Levels of packaging and packing required (see 5.1 and 5.2).
(d) When packing specified in 5.2.1.1 is required.

6.2 Selection of levels (military agencies). Based on the conditions known or expected to be encountered during shipment, handling, and storage of the specific item being procured, the contracting officer should select the appropriate level of pack in accordance with the criteria established in AR 700-15, NAVSUPINST PUB 4030.28, AFR 71-6, MCO 4030.14D or DSAR 4145.7, as applicable.

6.3 Packing specified in 5.2.1.1 is intended for transfer at sea operations or specific overseas requirements.

6.4 Supersession data. This specification includes the requirements of Interim Purchase Description S-12-1, dated 16 September 1971.

Custodians:

Army - GL
Navy - SA
Air Force - 45

Review activities:

Army - MD
Navy - MC, MS
DP - SS

Preparing activity:

Army - GL

CIVIL AGENCY COORDINATING ACTIVITIES:

USDA - AMS
VA - DMS
HEW - FDA
GSA - FSS

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DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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